Experiment 05

Title

Stock Price Prediction using Regression Models

SAP ID:	590013857
Dataset:	`https://www.kaggle.com/datasets/dgawlik/tesla-stock-price`

AIM

To develop and evaluate regression-based machine learning models for predicting stock prices using historical market data.

THEORY

Stock price prediction is a regression problem where the goal is to estimate future values based on historical data.

Key concepts:

- **Time Series Features**: Lag values, rolling averages, volatility
- **Regression Models**: Linear, Ridge, Lasso, SVR, Random Forest
- **Evaluation Metrics**: MAE, MSE, RMSE, R^2
- **Feature Engineering**: Improves predictive performance
- **Time-based Split**: Prevents data leakage

Commands/Code Used

```
# Core steps

# Load dataset
df = pd.read_csv("Tesla.csv")

# Feature engineering
df["Prev_Close"] = df["Close"].shift(1)
df["MA_5"] = df["Close"].rolling(5).mean()
df["MA_10"] = df["Close"].rolling(10).mean()

# Target variable
df["Target_Close"] = df["Close"].shift(-1)

# Train models
models = [
    LinearRegression(),
    Ridge(),
```

```
    Lasso(),
    SVR(),
    RandomForestRegressor()
]

# Evaluate using:
# MAE, MSE, RMSE, R2
```

Results

1. Stock Price Trend

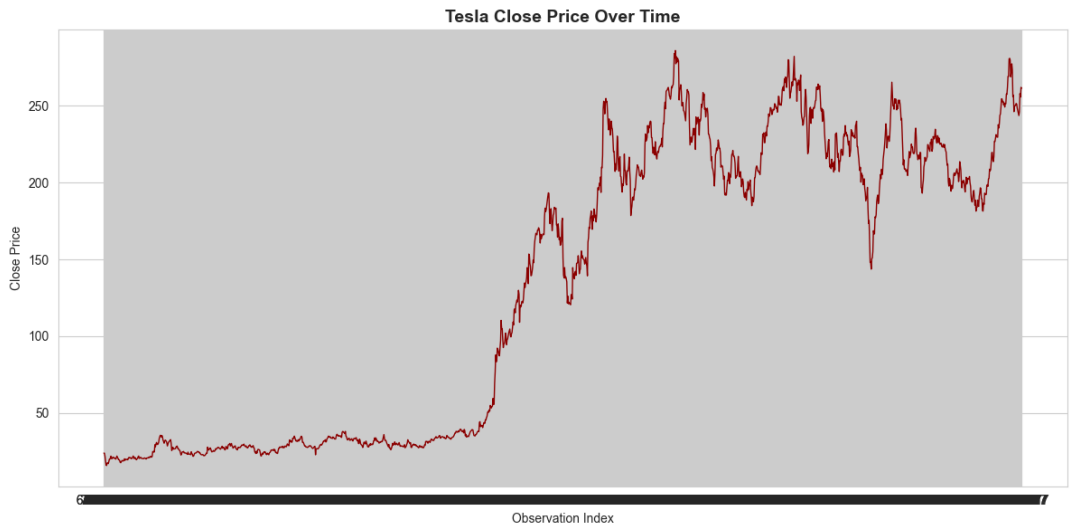

Figure 1: Tesla stock closing price over time

2. Correlation Heatmap

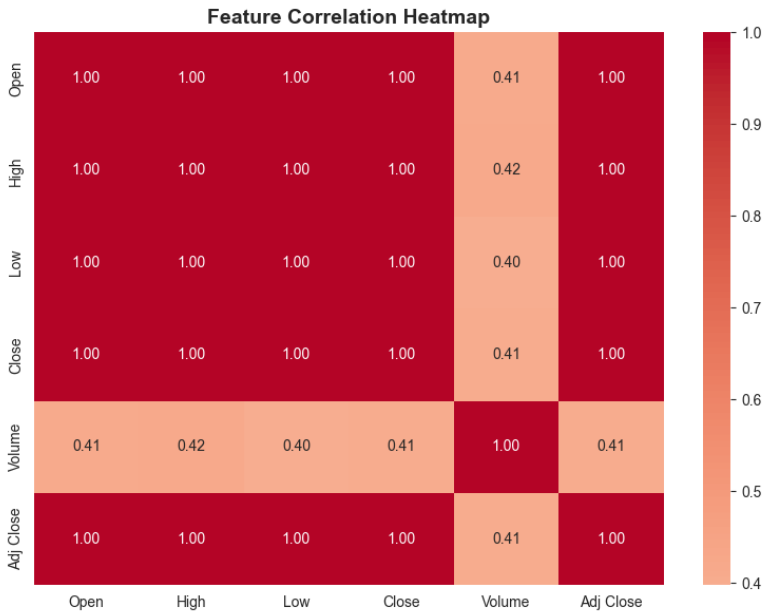

Figure 2: Feature correlation heatmap

3. Feature Importance

Figure 3: Top features influencing stock price

4. Model Comparison

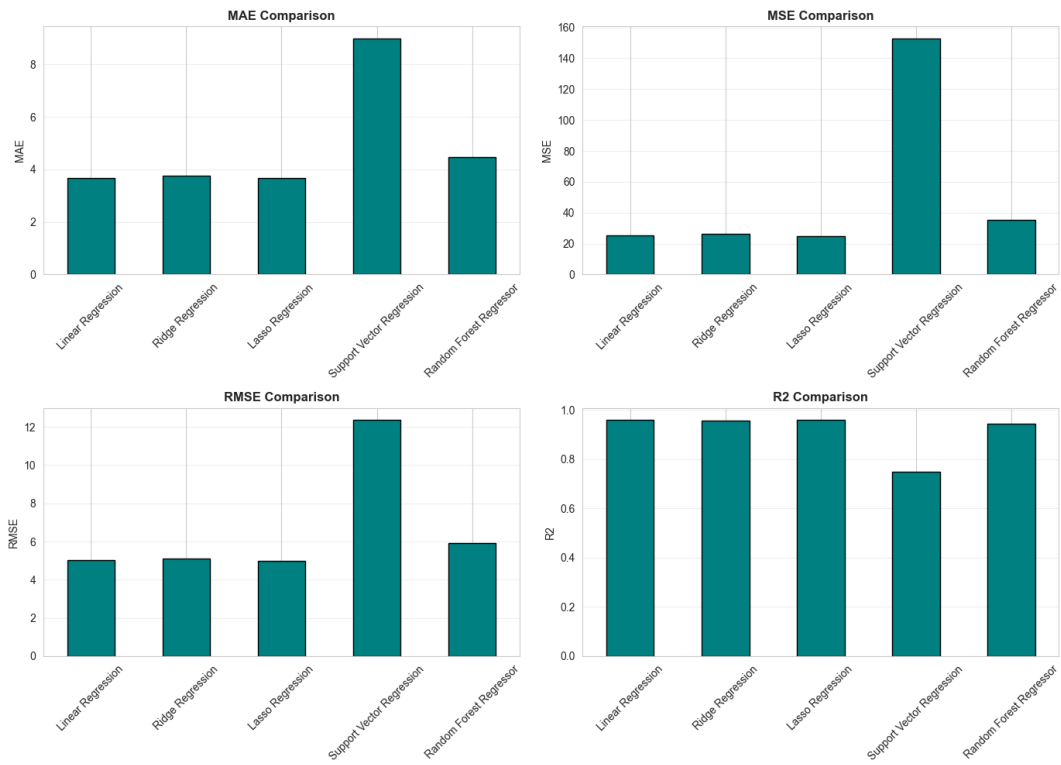

Figure 4: Comparison of regression models using MAE, MSE, RMSE, and R^2

5. Actual vs Predicted

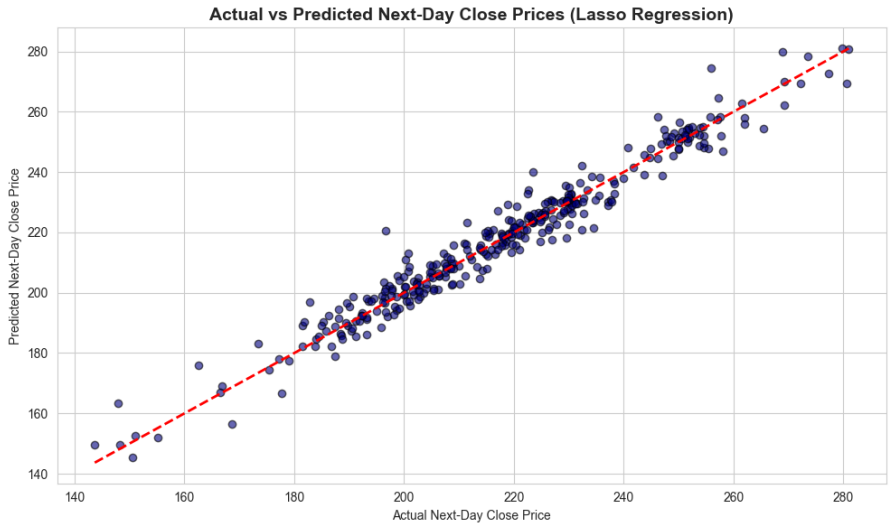

Figure 5: Predicted vs actual stock prices

6. Residual Analysis

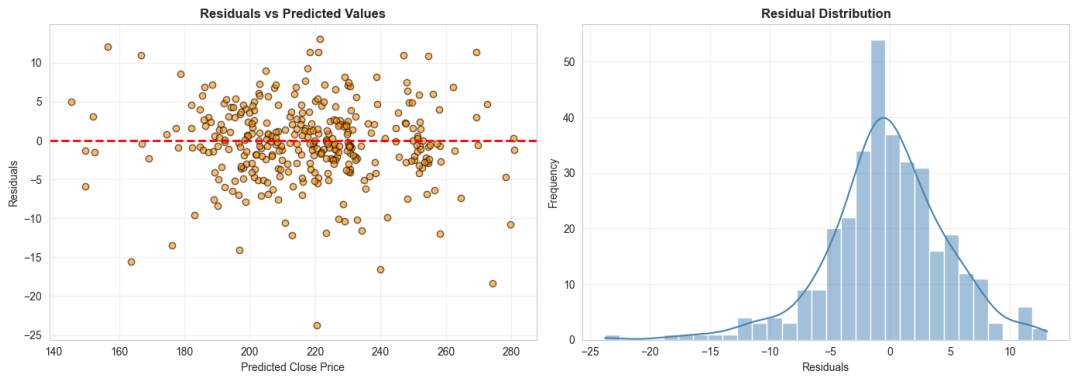

Figure 6: Residual analysis of predictions

7. Feature Relationships

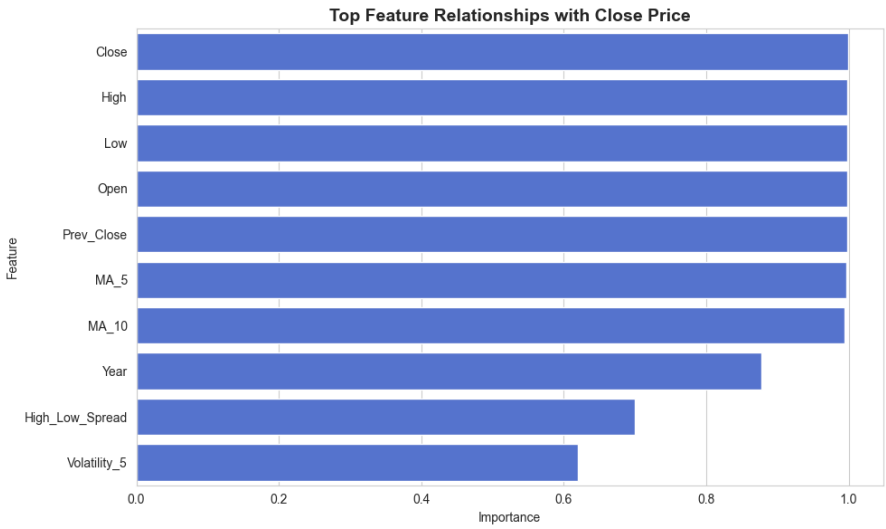

Figure 7: Feature relationships with close price

Conclusion

In this experiment, multiple regression models were applied to predict stock prices using historical data. Feature engineering techniques such as moving averages and lag values significantly improved model performance.

Among the models tested, regularized linear models and ensemble methods provided strong predictive results. Residual analysis confirmed that the model predictions were reasonably accurate.

This experiment highlights the importance of time-aware modeling, feature engineering, and evaluation techniques in financial data analysis.